

DRIVER BEHAVIOR AND SCORES ON THE MMPI¹

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If measurable personality variables are related to behavior of drivers, then persons responsible for training and licensing drivers should consider how this relationship can be used in order to reduce the number of highway accidents. Accidents result from a complex involving, in addition to the driver, the machine, the highway, and driving conditions such as the weather and the time of day. Any systematic programs that will effectively reduce the number of accidents must pay attention to all of these variables, but to date too little attention has been given to the personality of the driver.

In a study of traffic accident repeaters in Michigan and Connecticut (ENO, 1948), the Cornell Word Form, a type of personality test, distinguished to some extent between drivers with and without traffic accidents. Relationships were small. Harris (1950) found no significant differences between accident-prone and non-accident-prone employees using a multiple-choice Rorschach, the Rosenzweig Picture-Frustration Test, and 71 Bernreuter-type items. A military aviation study (O'Gorman & Kunkle, 1947) found some statistically significant and suggestive differences on the MMPI among three groups of pilots with different accident records. Of a no-accident group 12% had *Pd* standard scores of 60 or above and 2% *Ma* standard scores of 75 or above. Of a group with accidents due to violating rules, 20% had *Pd* scores of 60 or above and none had *Ma* scores of 75 or above, but of a group having accidents due to taxiing with carelessness, 16% had *Ma* scores of 75 or above.

The results that are available from studies

of relationships between personality and accidents are far from conclusive but are intriguing.

The purpose of this study was to determine, after a period of four to six years, if there is a relationship between the accident and violation records of drivers and their scores on the Minnesota Multiphasic Personality Inventory.

METHOD

The subjects of this study consisted of 993 residents of Minnesota who were registered as freshmen in the General College of the University of Minnesota in 1951, 1952, or 1953. This college offers a two-year program in general education, and the students in the college tend to be comparable in terms of ability and high school performance to the total group of high school graduates in the state. Most General College students come from the Minneapolis-St. Paul area, and most fall within the age range of 18 through 22. The mean age of the group was about 19 at the time the MMPI was administered.

During the three years, approximately 2,000 students were registered as freshmen in this college. Complete information regarding both personality inventory scores and driving records was available for slightly over 1,000, and as relatively few persons for whom such information was available were women, the study was restricted to men. The difficulty of obtaining driving records for women was due to the change of women's names resulting from marriage.

When a student enters the General College, he is expected to take a group of tests, including the MMPI. The proportion of entering students who actually take this test varies from year to year from 70 to 90%. The names of entering students were obtained, the students identified for whom test scores were available, and then these names were checked against the records of the Minnesota State Highway Department. From the driver's license application form in this department, the age, address, and date of issuing the driver's license were obtained. From the driver's record packet in this office was obtained information regarding violations, including date of violation, type of violation, place of violation, and department action taken against the driver, along with information about accidents which included date of accident, number of accidents, and the accident records file number.

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After the frequency distributions of the number of accidents and number of violations were obtained, three groups of 100 individuals each were selected from the 993 males for more intensive study. The first group consisted of the 100 persons with the greatest number of violations and accidents. In this group were persons who had five or more violations and/or three or more accidents. The low group consisted of 100 persons selected from the total group having no violations and no accidents on their records. A middle group of 100 persons was selected who fell at about the mean of each distribution, and these were people who had one and two violations combined with one accident. Frequency distributions were then tabulated, and the mean raw scores on the MMPI were compared for these three groups. The proportion of persons in the low group and the high group answering each question on the *Pd* (Psychopathic Deviate) scale and the *Ma* (Hypomanic) scale was determined, and the groups were compared in order to select items which differentiated between the groups.

A questionnaire was prepared and sent to each person in each of these three groups. Included on this questionnaire were questions asking for the number of miles driven per year, the number of years the person had been driving, the number of accidents the person had had, and the number of traffic tickets other than for parking. Two follow-ups to the original questionnaire resulted in a return of approximately 80%.

TABLE 1
FREQUENCY DISTRIBUTIONS OF NUMBER OF VIOLATIONS
AND NUMBER OF ACCIDENTS

Violations		Accidents	
Number of Violations	Number of Men	Number of Accidents	Number of Men
0	321	0	421
1	229	1	229
2	155	2	142
3	107	3	78
4	72	4	27
5	44	5	18
6	18	6	2
7	22	7	4
8	11	8	2
9	5	9	1
10	3		
11	2		
12	1		
13	1		
14	1		
15	1		
Total	993	Total	993

Note.—Mean violations = 1.88. Mean accidents = 1.09.

TABLE 2
MEAN RAW SCORE FOR EACH SCALE OF THE MMPI
FOR HIGH, MEDIUM, AND LOW GROUPS

Scale	Mean		
	Low Group	Middle Group	High Group
<i>?</i>	3.35	1.93	.42
<i>L</i>	3.77	3.41	3.27
<i>F</i>	4.58	4.91	5.81
<i>K</i>	15.03	14.82	14.75
<i>Hs</i>	12.40	13.08	12.56
<i>D</i>	17.31	18.54	17.71
<i>Hy</i>	18.70	19.73	19.24
<i>Pd</i>	22.52	23.29	23.99
<i>Mf</i>	22.65	22.58	21.70
<i>Pa</i>	8.50	8.98	9.11
<i>Pt</i>	28.97	29.53	27.28
<i>Sc</i>	28.71	28.60	27.99
<i>Ma</i>	20.47	20.57	21.44
<i>Si</i>	23.64	24.01	22.81

RESULTS

Table 1 presents the number of violations and the number of accidents for the 993 drivers as shown in the Highway Department records. Approximately one-third of the drivers had no violations, more than 10% had five or more violations. More than one-half of the drivers had at least one accident, and well over 10% had at least three accidents. The mean number of violations was 1.9, and the mean number of accidents was 1.1.

It must be noted here that many of these drivers had been driving for at least two or three years before they took the MMPI, and the number of violations and accidents includes all of those reported in Minnesota, some before and some after the time of testing.

Table 2 shows the mean scores on the scales for the three separate groups. In general, the means are not significantly different with the exception of the *Pd* and *Ma* scales, where the poor drivers tend to obtain elevated scores.

The correlations between the number of accidents and the number of violations and scores on the MMPI are shown in Table 3. With a group of 993 persons, a correlation of .08 is significant at the .01 level. Here it is seen that the significant correlations with *Pd*

TABLE 3
CORRELATION COEFFICIENTS FOR NUMBER OF VIOLATIONS AND NUMBER OF ACCIDENTS FOR EACH SCALE OF THE MMPI

Scale	Correlation Coefficient of Scale and Violations	Correlation Coefficient of Scale and Accidents
<i>L</i>	-.051	.007
<i>F</i>	.088	.079
<i>K</i>	.000	-.038
<i>Hs</i>	-.005	.000
<i>D</i>	-.029	-.007
<i>Hy</i>	.038	.002
<i>Pd</i>	.101	.083
<i>Mf</i>	-.040	-.032
<i>Pa</i>	.040	.046
<i>Pt</i>	-.004	-.027
<i>Sc</i>	.001	.000
<i>Ma</i>	.075	.097
<i>Si</i>	-.078	-.006

and *Ma* scales are so small that they have little practical value.

Examination of the profiles suggests that the obtained differences in the group analyses are small because the group of poor drivers contains a number of different kinds of profile groups or personality types. One might speculate, for instance, that one kind of poor driver, perhaps characterized by a high *Pd* score, is the driver who doesn't conform to much of anything, who doesn't have respect for the rights of other individuals or groups, who is quite completely self-centered. Another kind of driver well might be one who is extremely hostile toward other persons, perhaps as shown by a relatively high *Pa* score. His driving behavior may be motivated by a desire to get even with the next driver or to show the next fellow. A third type of driver might be one with an elevated *Ma* score, the driver who is in a great hurry and whose main interest is in reaching his destination as soon as possible, or sooner. When scores for these different types of drivers are combined, the elevated scores of one group tend to be compensated for in group analysis by the depressed scores of other groups, and in general the type of group analysis done here reveals little.

The self-reports of the number of years of driving and the number of miles driven per

year were compared for the three groups. A chi-square test showed that the poor driving group reported they had been driving a slightly significant greater number of years than the good driver group, but the difference in terms of number of miles per year was not significant. The number of accidents and violations in the self-reports were compared to those in the Highway Department records, and a strong and consistent tendency was found for drivers to report fewer accidents and violations. The mean number of accidents recorded for the poor drivers was 3.49, the mean number they actually reported themselves was 2.77, an understatement of 21%. The mean number of violations recorded was 5.88, and the mean of the personal reports was 3.98, an understatement of 33%.

CONCLUSION AND DISCUSSION

The results indicate that there is a small but observable relationship between scores on the MMPI and the number of accidents and violations of drivers. The observed differences suggest that a list of test items might provide some assistance in identifying drivers prone to have accidents because of their own personality characteristics.

If persons can be identified who have different types of personalities related to accidents and violations, perhaps the actual occurrence of the accident or violation can be reduced simply by calling to the attention of the individual the fact that he has this pattern of characteristics. Perhaps more intensive instruction would be necessary and it may be that certain persons can be identified, in driver training courses or through the means of driver's license examinations, who require more extensive counseling or perhaps even psychotherapy in order that they understand their own motivations sufficiently to control them when they are at the wheel.

Another possibility is suggested here by the difference in the self-reports and the actual recorded number of accidents and violations. Drivers who tend to under-report their violations may not be lying, they actually may be reporting themselves as they perceive themselves. Most persons perceive themselves as good, competent drivers, particularly young

men in this age range. Driving well is one of the things almost all young men feel themselves capable of doing. Any experience that tends to threaten this self-perception may arouse many defenses, and the most obvious kind of evidence that a person may not be as good a driver as he believes, that is, actual arrest or traffic tickets, may be quickly shut out from awareness or repressed so that the person can maintain the favorable perception he has of his own driving skill. The first problem with persons who have poor driving records may be to get them to realize they are poor drivers.

Adler (1941) proposed on the basis of a clinical study of 130 industrial accident cases five personality types prone to have repeated accidents. These were:

- Persons with revengeful attitudes
- Persons considering themselves as unlucky
- Persons longing to be pampered
- Persons who are overly ambitious
- Persons who are overly fearful

The concept of a personality type which is accident prone is difficult to defend, but the concept of habitual patterns of attitudes and behaviors which contribute differentially to the occurrence of accidents finds some support in this study.

An attitude of unconcern about others, with a failure to benefit from experience, accompanied by a disdain for rules and regulations, the *Pd* syndromes on the MMPI, well might be related to accidents.

Another set of attitudes more directly involves feelings of hostility. Persons who habitually respond to others with suspicion, who view every social situation as competitive, who strive for dominance in every instance, may as a result of these attitudes have accidents. This is the *Pa* syndrome.

Persons with high *Ma* scores may value effort and movement for their own sake, may pay little attention to social consequences of their acts, and may have accidents.

Not all persons with these attitudes will have accidents, for the results of some of these attitudes can be suppressed by other attitudes. The *Pa* syndrome may be suppressed by a tendency to withdraw from others, or by a fear of injury. None of these speculated relations are simple; rather, they are only suggestive.

SUMMARY

A group of 993 men was administered the MMPI when they were freshmen in college. From four to six years after this, their driving records were investigated, and a very small but statistically significant relationship was observed between scores on the *Pd* and *Ma* scales of the MMPI and the number of accidents and the number of violations of these individuals. The small size of these relationships might be due to the existence of more than one kind of personality pattern related to poor driving. Knowledge of the kind of personality organization and motivation of a driver may be useful for purposes of both licensing and training drivers.

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